



Official reprint from UpToDate®

www.uptodate.com © 2022 UpToDate, Inc. and/or its affiliates. All Rights Reserved.

Wolters Kluwer

Clinical features and diagnosis of Alzheimer disease

Authors: [David A Wolk, MD](#), [Bradford C Dickerson, MD](#)**Section Editor:** [Steven T DeKosky, MD, FAAN, FACP, FANA](#)**Deputy Editor:** [Janet L Wilterdink, MD](#)All topics are updated as new evidence becomes available and our [peer review process](#) is complete.**Literature review current through:** Sep 2022. | **This topic last updated:** Oct 08, 2021.

INTRODUCTION

Alzheimer disease (AD) is a neurodegenerative disorder of uncertain cause and pathogenesis that primarily affects older adults and is the most common cause of dementia [1]. The most essential and often earliest clinical manifestation of AD is selective memory impairment, although there are exceptions. While treatments are available that can ameliorate some symptoms of the illness, there is no cure and the disease inevitably progresses in all patients. The first approval by the US Food and Drug Administration (FDA) of a therapy that is potentially disease-modifying provides a mandate for a specific diagnosis of AD in patients with cognitive impairment and dementia.

This topic reviews the clinical manifestations and diagnosis of AD. Other topics review the risk factors and treatment of AD:

- (See "[Epidemiology, pathology, and pathogenesis of Alzheimer disease](#)".)
- (See "[Treatment of Alzheimer disease](#)".)

Other topics review the approach to patients with cognitive impairment and dementia and the clinical features of other dementia syndromes:

- (See "[Evaluation of cognitive impairment and dementia](#)".)
- (See "[Mild cognitive impairment: Epidemiology, pathology, and clinical assessment](#)".)
- (See "[Etiology, clinical manifestations, and diagnosis of vascular dementia](#)".)
- (See "[Clinical features and diagnosis of dementia with Lewy bodies](#)".)
- (See "[Frontotemporal dementia: Clinical features and diagnosis](#)".)

CLINICAL FEATURES

Age of onset

- **Typical AD** – AD is characteristically a disease of older age [2]. It is unusual for AD to occur before age 60. The incidence and prevalence of AD increase exponentially with age, essentially doubling in prevalence every 5 years after the age of 65 years. (See ["Epidemiology, pathology, and pathogenesis of Alzheimer disease", section on 'Incidence and prevalence'](#).)
- **Early-onset AD** – Onset of symptoms before 65 years of age is uncommon, occurring in approximately five percent of patients with AD. Such patients often present for evaluation due to concerns about job performance. Many of these patients have no clear familial pattern and thus would be considered sporadic. Some exhibit familial clustering, but usually with affected family members being older at symptom onset.

People with early-onset AD often present with symptoms somewhat atypical for this disease, such as language, visual, or mood-behavioral changes rather than predominant memory loss.

- **Inherited forms of AD** – These rare forms of AD routinely present before 65 years of age, and frequently in the fifth decade or earlier. These account for less than 1 percent of all cases of AD. They typically exhibit an autosomal-dominant inheritance pattern related to mutations in genes that alter beta-amyloid ($A\beta$) protein production or metabolism, including amyloid precursor protein (*APP*), presenilin-1 (*PSEN1*), and presenilin-2 (*PSEN2*).

In a meta-analysis on 1307 patients with autosomal-dominant AD, the mean age of symptom onset was 46 years and was highly correlated with parental age of onset and mutation type [3]. Patients with *PSEN1* mutations had the earliest median age of onset (43 years) ( figure 1). The range of symptom onset across all mutation types is nonetheless fairly broad, with some presentations in the fourth decade and some mutations not manifesting symptoms until the seventh decade (see ["Early-onset dementia in adults", section on 'Neurodegenerative dementias'](#)). If this is suspected, it is important to collect detailed information about family history to document a potential autosomal-dominant familial pattern.

- **AD associated with Down syndrome** – Individuals with Down syndrome have an additional gene dose of *APP* due to trisomy of chromosome 21 and inevitably develop AD

pathology. Symptoms emerge at an earlier age, 10 to 20 years younger than the general population with AD [4]. (See "[Genetics of Alzheimer disease](#)", section on 'Trisomy 21'.)

Cardinal symptoms

Memory impairment — Memory impairment is the most common initial symptom of AD dementia. Even when not the primary complaint, memory deficits can be elicited in most patients with AD at the time of presentation. Deficits in other cognitive domains may appear with or after the development of memory impairment.

The pattern of memory impairment in AD is distinctive [5]. Declarative episodic memory (memory of events occurring at a particular time and place) is usually profoundly affected in AD. This type of memory depends heavily on the hippocampus and other medial temporal lobe structures. By contrast, subcortical systems supporting procedural memory and motor learning are relatively spared until quite late in the disease. Memory for facts such as vocabulary and concepts (semantic memory) often becomes impaired somewhat later. Semantic memory is supported by neocortical temporal regions, particularly in the anterior temporal lobe. Subtler impairments of semantic memory may be a relatively early feature given early temporal lobe pathology [6].

Within episodic memory, there is a distinction between immediate recall (eg, mental rehearsal of a phone number), memory for recent events (which comes into play once material that has departed from consciousness must be recalled), and memory of more distant events (remote memory). Memory for recent events, served by the hippocampus, entorhinal cortex, and related structures in the medial temporal lobe, is prominently impaired in early AD [7-9]. By contrast, immediate memory (encoded in the sensory association and prefrontal cortices) is spared early on, as are memories that are consolidated for long periods of time (years), which can be recalled without hippocampal function.

The early memory deficit in AD is most precisely described as anterograde long-term episodic amnesia. Long-term memory can encompass time frames from less than a minute to years, but because the absolute time interval over which long-term memory can fail may actually be short (eg, inability to recall a few words after a couple minutes of distraction), patients and caregivers typically describe "short-term memory" problems. For this reason, we try to avoid the confusion afforded by the technical terms of long-term and short-term memory and use the term "recent memory impairment" to refer to the characteristic impairment.

Memory deficits develop insidiously and progress slowly over time, evolving to include deficits of semantic memory and immediate recall. Impairments of procedural memory appear only in late stages of AD.

Memory is usually tested by asking patients to learn and recall a series of words or objects immediately and then at a delay of 5 to 10 minutes. Impaired ability to recall objects with selective cues (hints) or recognize items as previously studied on a recognition memory test represents a more severe deficit and one that may be particularly specific for AD in its early presentation, as it reflects specific involvement of medial temporal lobe function [10]. Questions about orientation and recent current events are also useful memory tests. (See ["The mental status examination in adults", section on 'Memory'](#).)

Clinicians should compare an informant's report with the patient's report of symptoms of memory loss in daily life, recognizing that many older individuals are unreliable reporters of their own memory impairment and can both overestimate and underestimate their deficits [11]. In addition, lack of insight may cause patients to deny or under-report their symptoms. It is critical to obtain the perspective of someone who knows the patient well.

Executive function and judgment/problem solving — In early stages of AD, impairment in executive function may range from subtle to prominent [12-14]. In many cases, the patient underreports these symptoms, and an informant interview is critical. Family members and coworkers may find the patient less organized or less motivated; multitasking is often particularly compromised. As the disease progresses, an inability to complete tasks typically emerges.

Reduced insight into deficits (anosognosia) is a common but variable feature of AD [15,16]. It is common for patients to underestimate their deficits and offer alibis or explanations for them when they are pointed out. Interviewing an informant who has known the patient over time (usually a family member) is necessary; often it is the family member, not the patient, who brings the complaint of cognitive impairment to medical attention.

Loss of insight increases over time along with overall disease severity [17]. Such loss of insight may be associated with behavioral disturbances. Those with relatively preserved insight are more likely to be depressed; those with more impaired insight are likely to be agitated, be disinhibited, and exhibit psychotic features [18,19]. Lack of insight also may impact safety, as patients may attempt to take on tasks that they no longer have the capacity to perform effectively (eg, driving). (See ["Management of the patient with dementia", section on 'Driving'](#).)

Impairments in other cognitive domains — Other cognitive domains become affected in patients with AD. Visuospatial impairments are often present relatively early, while deficits in language often manifest later in the disease course. These deficits develop and progress insidiously. Less commonly, language deficits, visuospatial abnormalities, or even executive

functions may be impaired as the most prominent initial symptom [12,13]. (See '[Atypical presentations](#)' below.)

Behavioral and psychologic symptoms — Neuropsychiatric symptoms are common in AD, particularly in the middle and late course of disease. These can begin with relatively subtle symptoms including apathy, social disengagement, and irritability. Apathy can be difficult to distinguish from depression, which can be difficult to diagnose in the setting of dementia, and the differences should be sought diligently because of treatment implications. In some cases empiric treatment of presumed depression is a reasonable management decision. (See "[Management of neuropsychiatric symptoms of dementia](#)", section on '[Depression](#)'.)

More problematic in patient management is the emergence of behavioral disturbances, including agitation, aggression, wandering, and psychosis (hallucinations, delusions, misidentification syndromes). A concomitant medical illness, medication toxicity, and other causes of delirium should be considered whenever new behavioral disturbances arise, especially if acute or subacute. The behavioral disturbances associated with AD are discussed in detail separately. (See "[Management of neuropsychiatric symptoms of dementia](#)".)

Other signs and symptoms

Apraxia — Dyspraxia, or difficulty performing learned motor tasks, usually occurs later in the disease after deficits in memory and language are apparent [20]. Before it is clinically manifest, dyspraxia can be elicited by asking the patient to perform ideomotor tasks, ie, pantomime the use of tools (eg, "show me how you would use a comb") [21,22]. Clinical dyspraxia leads to progressive difficulty first with complex, multistep motor activities, then with dressing, using utensils to eat, and other self-care tasks, and is a significant contributor to dependency in mid- to late stages of AD [23].

Olfactory dysfunction — Changes in olfactory function are common in patients with AD and have been explored as a diagnostic tool. However, the predictive value of simple odor detection testing is limited [24,25], and the challenges of standardized assessment have limited widespread use of olfactory assessment [26,27]. Furthermore, olfactory dysfunction is not a clinical symptom that is often noted by patients or their families. That said, it may enhance classification of patients beyond cognitive measures alone [28].

Sleep disturbances — Sleep disturbances are common in patients with AD [29]. Patients with AD spend more time in bed awake and have more fragmented sleep compared with older adults without AD. Such changes may occur very early in the disease process, including in patients who are cognitively normal but have biomarker evidence of A β deposition [30-32]. (See

["Sleep-wake disturbances and sleep disorders in patients with dementia", section on 'Sleep changes in aging and dementia'.\)](#)

Seizures — Seizures occur in 10 to 20 percent of patients with AD, usually in the later stages of disease [33-37]. Younger patients, including those with autosomal-dominant forms of AD, may be at higher risk for seizures, which may occur early in the course of disease [38,39].

The predominant seizure type is focal nonmotor with impaired awareness, with symptoms often suggestive of medial temporal lobe onset (eg, amnesic spells, unexplained emotions, metallic taste, rising epigastric sensation) [39]. (See ["Seizures and epilepsy in older adults: Etiology, clinical presentation, and diagnosis", section on 'Epilepsy \(unprovoked seizures\)'.](#))

Motor signs — In the early stages, patients with AD generally have a normal neurologic examination except for the cognitive examination. While pyramidal and extrapyramidal motor signs, myoclonus, and seizures do occur in patients with AD, these are typically late-stage findings [40]. If these are clinically apparent in early to middle stages, alternative diagnoses should be considered [41].

Myoclonus may emerge in some patients with AD, typically those with more rapid than usual decline. Similarly, primitive reflexes (grasp, snout reflexes, gegenhalten) and incontinence are late, rather than early, features of AD.

Atypical presentations — A minority of patients with AD do not present in the classic fashion with progressive amnesic dementia [42,43]. Distinguishing these cases from other causes of dementia can be challenging. Neuropathologic studies in such patients have found Alzheimer pathology with an atypical neuroanatomic distribution (particularly neurofibrillary tangles) that corresponds with the anatomy of the most prominent clinical deficits; neuroimaging studies often have corresponding findings [43-48]. Some case series suggest that these atypical forms are more common in early-onset presentations (before 65 years of age) in patients who do not carry one of the autosomal-dominant familial mutations [49,50].

- **Posterior cortical atrophy** – This syndrome manifests with progressive cortical visual impairment [42,46,51-56]. Patients are often first evaluated by optometrists or ophthalmologists for visual complaints, such as difficulty reading and driving [43,54]. A proposed consensus classification framework requires three or more of the following early or presenting features [57]:
 - Space perception deficit
 - Simultanagnosia (ie, the inability to integrate a visual scene despite adequate visual acuity to resolve individual elements)

- Object perception deficit
- Constructional dyspraxia
- Environmental agnosia
- Oculomotor apraxia (the inability to direct gaze accurately to a new target)
- Dressing apraxia
- Optic ataxia (the inability to reach accurately under visual guidance)
- Alexia
- Left/right disorientation
- Acalculia
- Limb apraxia
- Apperceptive prosopagnosia
- Agraphia
- Homonymous visual field defect
- Finger agnosia

In addition, there should be relative sparing of anterograde memory function, speech and nonvisual language functions, executive functions, and behavior and personality.

Neuroimaging usually shows predominant occipitoparietal or occipitotemporal atrophy, hypometabolism, or hypoperfusion [45,57-60].

Some patients with a biparietal variant present with dyspraxia and have difficulty completing simple bimanual tasks, such as dressing [42,61]. Other early clinical features can include visuospatial disorientation, dysgraphia, and language impairment with semantic memory deficits (see '[Memory impairment](#)' above). Neuropathologic examination and neuroimaging studies demonstrate prominent involvement of the parietal lobes bilaterally.

In most patients, neuropathologic examination reveals Alzheimer pathology with a greater predominance of pathology involving visual association areas and even primary visual cortex compared with more typical presentations [43,55,59,62]. A minority of patients with this syndrome have alternative pathologies, such as dementia with Lewy bodies (DLB), corticobasal degeneration (CBD), prion disease, or frontotemporal lobar degeneration (FTLD) [55,59,63,64]. In such patients, other core features of the respective diseases are usually present (eg, fluctuating cognition, recurrent visual hallucinations, and parkinsonism in DLB; limb rigidity, akinesia, dystonia, or myoclonus in CBD) [57]. (See '[Clinical features and diagnosis of dementia with Lewy bodies](#)' and '[Corticobasal degeneration](#)', section on 'Clinical features' and '[Diseases of the central nervous system](#)'.)

caused by prions" and "Frontotemporal dementia: Epidemiology, pathology, and pathogenesis".)

- **Primary progressive aphasia** – Primary progressive aphasia (PPA) refers to a clinically and pathologically heterogeneous group of neurodegenerative disorders characterized by progressive language difficulty with relative sparing of memory and other cognitive functions, at least early in the disease course. Based on the type of language impairment, PPA is further subclassified into three variants: nonfluent, semantic, or logopenic. PPA is usually associated with FTLD rather than Alzheimer pathology. However, a significant percentage (up to one-third) are found at autopsy to have AD instead [42,43,45,65-67]. The clinical variant of PPA most likely to be due to AD is the logopenic variant, characterized by frequent word-finding pauses and paraphasic speech errors without major grammar or comprehension deficits [45,67-71]. AD pathology is also sometimes found in patients with nonfluent or semantic variants of PPA, but this is less common [42,43]. (See "Frontotemporal dementia: Clinical features and diagnosis".)

In patients with logopenic variant PPA, structural imaging often shows predominant left posterior perisylvian or parietal atrophy [72]. 18-F fluorodeoxyglucose positron emission tomography (FDG-PET) and single-photon emission computed tomography (SPECT) may highlight hypometabolism or hypoperfusion in the same areas. Elevated signal with amyloid PET imaging suggests a role for AD pathology and is seen in approximately 85 percent of logopenic variant cases and in approximately 20 percent of semantic and nonfluent variants in one series [73], but some cases have mixed pathologies [74], so amyloid PET should not be viewed as diagnostic of AD.

- **Dysexecutive or "frontal" variant** – A subset of patients with AD present with prominent deficits in executive function relative to amnesia. In one case series of 88 such patients, magnetic resonance imaging (MRI) revealed more thinning in the frontoparietal cortical regions, the clinical course appeared to involve more rapid progression, and the prevalence of the apolipoprotein E (*APOE*) ϵ 4 allele was lower compared with the more typical AD patients with prominent memory deficits [75,76].

Mixed dementias — It is common for Alzheimer pathology to coexist with other processes, including vascular lesions, cortical Lewy bodies, TAR DNA binding protein 43 (TDP-43) deposits, argyrophilic grain disease, and Parkinson disease. The combination of two pathologies can influence the clinical presentation and course of the disease and present diagnostic challenges [77]. In general, these additional pathologies result in greater likelihood of dementia and faster than usual rate of decline [78-80].

As expected, the most common combination is that of AD and vascular dementia. This is discussed in detail separately. (See ["Etiology, clinical manifestations, and diagnosis of vascular dementia"](#).)

Another combination that is described is that of diffuse Lewy body and AD dementia [81-83]. In affected patients, prominent memory loss can coexist with one or more features of DLB, including visual hallucinations, fluctuations in sensorium, parkinsonism, and falls. (See ["Clinical features and diagnosis of dementia with Lewy bodies"](#).)

CLINICAL COURSE

AD progresses inexorably. The progress of the disease can be measured with mental status scales such as the Mini-Mental State Examination (MMSE), the Montreal Cognitive Assessment (MoCA), the clinical dementia rating (CDR) scale, and/or an instrument measuring independent daily function such as the Functional Activities Questionnaire (📄 [table 1](#)). While the clinical course as measured by such scales is not necessarily linear, a number of studies have found that patients decline 3 to 3.5 points on average on the MMSE each year [84-87], with a minority (<10 percent) having a more rapidly progressive decline of 5 to 6 points on annual MMSE [88]. (See ["Mental status scales to evaluate cognition"](#), section on 'Longitudinal assessment'.)

An older age of onset of AD (>80 years) may be associated with a slower rate of decline compared with younger patients [89]. Conversely, early neuropsychiatric symptoms including psychosis, agitation, and aggression have been associated with more rapid decline [90].

The average life expectancy after a diagnosis of AD has been reported to be between 8 and 10 years but may range from 3 to 20 years, and it depends heavily on how impaired the person is at the time of diagnosis [91-94]. Survival also relates to age at onset of symptoms. Patients generally succumb to terminal-stage complications that relate to advanced debilitation, such as dehydration, malnutrition, and infection. (See ["Care of patients with advanced dementia"](#).)

EVALUATION

Clinical assessment — AD should be suspected in any older adult with insidious onset, progressive decline in memory, and at least one other cognitive domain leading to impaired functioning. Thus, a detailed cognitive and general neurologic examination is paramount. Clinicians should also consider potential contributors to the dementia syndrome such as adverse effects of medication, depression, and metabolic disorders and deficiencies. (See ["Evaluation of cognitive impairment and dementia"](#), section on 'Evaluation'.)

Many clinicians make use of standardized mental status scales to document the presence and progression of dementia. The Montreal Cognitive Assessment (MoCA) [95,96] is our preferred brief assessment tool because it has better sensitivity for executive and language dysfunction compared with other brief tests such as the Mini-Mental State Examination (MMSE). The typical cutoff score for normal performance on the MoCA is 26 (ie, 25 and below is considered abnormal). Cutoffs should be adjusted based on appropriate norms, including education adjustment [97]. The MoCA is available online and in several languages at www.mocatest.org. (See "[Mental status scales to evaluate cognition](#)", section on '[Montreal Cognitive Assessment \(MoCA\)](#)'.)

The diagnosis of dementia cannot be made simply on the basis of a low score on one of these tests; the most important part of the diagnostic evaluation is a detailed history including the perspective of an informant (eg, spouse or adult child), interviewed separately from the patient if possible. The use of validated questionnaires to document the preservation or loss of independent function, as well as the presence or types of neuropsychiatric symptoms, is also an important part of the evaluation. Validated questionnaires to measure cognitive symptoms also can be helpful [98]. Additional detail on these instruments is presented separately. (See "[Mental status scales to evaluate cognition](#)".)

Role of neuropsychologic testing — Formal neuropsychologic assessment can be helpful in the evaluation of individuals with cognitive impairment and dementia. Cognitive testing under standardized conditions using demographically appropriate norms is more sensitive to the presence of impairments, especially impairments of executive function. Neuropsychologic assessment can be useful in a variety of settings:

- To establish a baseline in order to follow the patient over time. (See "[Evaluation of cognitive impairment and dementia](#)", section on '[Neuropsychological testing](#)'.)
- To help differentiate among different forms of neurodegenerative dementias or between a neurodegenerative dementia and other etiologies of cognitive impairment, such as cerebrovascular disease or depression. (See "[Evaluation of cognitive impairment and dementia](#)", section on '[Neuropsychological testing](#)'.)
- To assess competencies and guide recommendations pertaining to driving, financial decisions, and need for increasing supervision. (See "[Management of the patient with dementia](#)".)
- To identify opportunities for compensatory or rehabilitative treatment strategies. (See "[Management of the patient with dementia](#)", section on '[Rehabilitation](#)'.)

Neuroimaging — Brain imaging, preferably with magnetic resonance imaging (MRI), is indicated in the evaluation of patients with suspected AD [99]. Brain MRI can document potential alternative or additional diagnoses including cerebrovascular disease, other structural diseases (chronic subdural hematoma, cerebral neoplasm, normal pressure hydrocephalus), and regional brain atrophy suggesting frontotemporal dementia (FTD) or other types of neurodegenerative disease. (See ["Evaluation of cognitive impairment and dementia"](#), section on 'Neuroimaging'.)

- **MRI** – Structural MRI findings in AD include both generalized and focal atrophy, as well as white matter lesions. In general, these findings are nonspecific.

The most characteristic focal finding in AD is reduced hippocampal volume or medial temporal lobe atrophy [47,100-103]. Because hippocampal volumes decline in normal aging, however, age-specific criteria are needed [100,101,104]. The finding of hippocampal atrophy provides incremental support for a diagnosis of AD in a patient with a typical clinical presentation, but it is not sufficiently specific to contribute significantly to the accuracy of the diagnosis over the clinical assessment alone [105]. Some studies have suggested that MRI features may predict rate of decline of AD and in the future may guide treatment decisions [85,106]. Hippocampal volumetry using age-corrected norms available from the Alzheimer Disease Neuroimaging Initiative can predict rates of progression of mild cognitive impairment (MCI) to dementia [107]. However, the tools to generate these measurements are not in wide use, nor have these findings been validated in a clinical practice setting.

- **FDG-PET and SPECT** – Functional brain imaging with 18-F fluorodeoxyglucose positron emission tomography (FDG-PET) or single-photon emission computed tomography (SPECT) reveals distinct regions of hypometabolism (PET) and hypoperfusion (SPECT) in AD. These areas include the hippocampus, the precuneus (mesial parietal lobes), and the lateral parietal and posterior temporal cortex [9,108-113]. In practice, FDG-PET may be most useful in distinguishing AD from FTD in patients with atypical presentations [114-117], as well as discriminating from non-neurodegenerative conditions, such as depression. FDG-PET and SPECT are the only functional neuroimaging methods that are currently reasonably widely available for clinical use.
- **Amyloid PET imaging** – Amyloid PET tracers ([florbetapir F-18](#), [flutemetamol F-18](#), [florbetaben F-18](#)) measure amyloid lesion burden in the brain; these aid in the diagnosis of AD, differentiating AD from other causes of dementia [117-140]. A large study of the use of amyloid PET imaging in clinical decision-making was completed in December 2017 ([NCT02420756](#)) and revealed that an amyloid PET result significantly impacted clinical

management and etiologic diagnosis [141]. At present, these scans are generally not covered by most health insurances plans in the United States.

These tracers have been approved by regulatory agencies in the United States and elsewhere as qualitative assessments of beta-amyloid (A β) plaque density [142,143]. Since there are issues with how much ligand binding to plaques constitutes a "positive" scan, the US Food and Drug Administration (FDA) approval specifies that an amyloid PET scan that is negative decreases the likelihood that a patient with dementia has AD. In a symptomatic dementia patient, a positive scan indicates that the person has amyloid plaque pathology, but such a finding does not rule out a coexisting pathology. Based on the FDA approval label, scans are determined positive or negative based on a radiologist's clinical read, although there are more quantitative approaches that are used in research and may complement such clinical assessments.

A consensus opinion of the Amyloid Imaging Task Force, the Society of Nuclear Medicine, and the Alzheimer's Association concluded that amyloid imaging is not appropriate in patients who meet the core clinical criteria for probable AD and have a typical age of onset, and such a scan should not be used to determine dementia severity [144]. These guidelines may be adjusted with the approval of [aducanumab](#), for which a positive amyloid scan is generally believed to be a prerequisite [145]. (See "[Treatment of Alzheimer disease](#)", section on 'Aducanumab'.)

- **Tau PET imaging** – A number of tau PET imaging tracers have been in development and hold potential as markers of the tauopathy of AD. A number of studies have demonstrated that these tracers better track disease progression and more highly associate with patterns of atrophy and clinical features than amyloid PET does [146,147].

One such tracer, [flortaucipir F-18](#), was approved by the FDA for determining the burden of neurofibrillary tangles in individuals being evaluated for AD; this tracer is not useful for measuring non-AD tauopathies such as frontotemporal lobar degeneration (FTLD) [148,149]. As with amyloid PET, there is limited insurance coverage of this agent.

Role of biomarkers — There are several widely investigated biomarkers for the molecular and degenerative process of AD that can be supportive of a diagnosis of AD but are not yet recommended for routine diagnostic purposes ([table 2](#)) [12]. Such testing can add incremental confidence to a clinical diagnosis of AD, however, and can be useful in certain circumstances, including early-onset dementia and atypical presentations of AD in which the differential diagnosis includes other non-amyloid neurodegenerative diseases such as FTD. Evidence of the presence of amyloid is also likely to guide initiation of [aducanumab](#), which

specifically targets this protein. (See "[Treatment of Alzheimer disease](#)", section on '[Aducanumab](#)'.)

Molecular biomarkers of A β protein deposition include:

- Low cerebrospinal fluid (CSF) A β 42 (or A β 42:A β 40 ratio)
- Positive amyloid PET imaging using one of the amyloid PET tracers

Appropriate use guidance states that one of these tests should be positive in patients who are treated with [aducanumab](#). (See "[Treatment of Alzheimer disease](#)", section on '[Aducanumab](#)'.)

Biomarkers of tau deposition (a key component of neurofibrillary tangles) include:

- Increased CSF total tau and phospho-tau
- Tau PET imaging using [flortaucipir F-18](#)

In addition to these molecular biomarkers, there are several topographic or neurodegenerative biomarkers used to assess the downstream brain changes that correlate with the regional distribution of neuronal dysfunction and eventual neuronal death associated with AD [150]. These include medial temporal lobe atrophy on MRI and reduced glucose metabolism in the temporoparietal regions on FDG-PET. (See '[Neuroimaging](#)' above.)

In general, the topographic biomarkers are less specific than the molecular biomarkers but correlate better with the emergence of clinical symptoms. None of these tests is valid as a standalone diagnostic test and their combination may best predict outcomes [151]. Research criteria have begun to incorporate these biomarkers in the definition of AD with the goal of providing a biologically based diagnosis rather than a purely clinical diagnosis [152]. These criteria define AD based on the presence of A β and tau biomarkers and are irrespective of the patient's clinical impairments. Markers of neurodegeneration also provide information about disease staging, and, taken together, this has been referred to as the amyloid, tau, and neurodegeneration (ATN) framework.

Plasma biomarkers show promise but do not currently have an established role in clinical practice; more studies are needed [153-156]. Decreased APOE and APOE ϵ 4 plasma levels as well as a variety of other plasma/serum and CSF proteins have been assessed for predictive value for AD in persons without dementia and in patients with MCI [157-160]. Ongoing research is investigating the role of such biomarkers that may help distinguish AD from other forms of dementia [161,162]. In particular, measures of plasma phospho-tau₁₈₁ and phospho-tau₂₁₇ have demonstrated strong correlation with CSF phospho-tau measures and association with amyloid and tau PET imaging [161-163]. Similarly, a plasma measure of β 42/ β 40 using an

immunoprecipitation and liquid chromatography-mass spectrometry assay has shown promise in detection of amyloid [164] and can be commercially ordered, although it is not currently covered by insurance payers in the United States.

Studies of these and other emerging plasma/serum and CSF biomarkers for AD in the assessment of prognosis are discussed in more detail separately. (See "[Mild cognitive impairment: Prognosis and treatment](#)", section on 'CSF biomarkers' and "[Mild cognitive impairment: Prognosis and treatment](#)", section on 'Plasma biomarkers'.)

Other laboratory testing — Routine laboratory tests are not useful in the positive diagnosis of AD; however, some laboratory tests are indicated to exclude contributing secondary causes or conditions that may exacerbate cognitive impairment. (See "[Evaluation of cognitive impairment and dementia](#)", section on 'Laboratory testing'.)

Genetic testing — Genetic testing is not recommended in the routine evaluation of patients with AD. *APOE* genotyping adds marginally to the predictive value of clinical criteria for AD and may stratify risk of progression of amnesic MCI to AD, but both false positives and negatives occur [165,166].

Genetic testing for *APP*, *PSEN1*, and *PSEN2* mutations is commercially available but should be reserved for cases in which young-onset dementia occurs in the setting of a family history positive for an autosomal-dominant distribution of early-onset cases. This should not be performed in asymptomatic family members or patients without appropriate professional genetic counseling, since there would be implications for family members as well as the patient. (See "[Genetics of Alzheimer disease](#)", section on 'Genetic testing'.)

DIAGNOSIS

A detailed clinical assessment provides reasonable diagnostic accuracy for AD in the majority of patients, although sensitivity and specificity in particular are limited [12]. That said, misdiagnosis is almost always another neurodegenerative or nonreversible cause. The clinical criteria for AD include a history of insidious onset and progressive course of cognitive decline, exclusion of other etiologies, and documentation of cognitive impairments in one or more domains.

Definitive diagnosis of AD requires histopathologic examination, which is rarely done in life [167-169]. The diagnosis of AD in practice depends on the clinical criteria outlined below.

Alzheimer disease dementia — Criteria for the diagnosis of probable AD dementia have been established by the National Institute on Aging and the Alzheimer's Association (NIA-AA) and most recently updated in 2011 [12,41].

- **NIA-AA criteria for probable AD dementia** require the presence of dementia **and** the following characteristics:
 - Interference with ability to function at work or at usual activities
 - A decline from a previous level of functioning and performing
 - Not explained by delirium or major psychiatric disorder
 - Cognitive impairment established by history-taking from the patient **and** a knowledgeable informant; and objective bedside mental status examination or neuropsychologic testing
 - Cognitive impairment involving a minimum of two of the following domains:
 - Impaired ability to acquire and remember new information
 - Impaired reasoning and handling of complex tasks, poor judgment
 - Impaired visuospatial abilities
 - Impaired language functions
 - Changes in personality, behavior, or comportsment
 - Other core clinical criteria include:
 - Insidious onset
 - Clear-cut history of worsening
 - Initial and most prominent cognitive deficits are one of the following: amnestic presentation (ie, impairment in learning and recall of recently learned information) **or** nonamnestic presentations (including either a language presentation, with prominent word-finding deficits; a visuospatial presentation, with visual cognitive deficits; or a dysexecutive presentation, with prominent impairment of reasoning, judgment, and/or problem solving)

- No evidence of substantial concomitant cerebrovascular disease, core features of dementia with Lewy bodies (DLB), prominent features of behavioral variant frontotemporal dementia (FTD) or prominent features of semantic or nonfluent/agrammatic variants of primary progressive aphasia (PPA), or evidence of another concurrent, active neurologic or non-neurologic disease or use of medication that could have a substantial effect on cognition
- **NIA-AA criteria for possible AD dementia** include either of the following clinical scenarios [12]:
 - Atypical course – The core clinical criteria above are met in terms of the nature of the cognitive deficits, but there is either a sudden onset of cognitive impairment or insufficient historical detail or objective documentation of progressive decline.
 - Etiologically mixed presentation – All of the core clinical criteria for AD dementia are met, but the individual also has evidence of concomitant cerebrovascular disease, features of DLB other than the dementia itself, or evidence for another neurologic or medical comorbidity or medication that could have a substantial effect on cognition.
- **The Diagnostic and Statistical Manual of Mental Disorders (DSM) criteria for AD** are also commonly used. Criteria for AD were revised in 2013 (table 3) [170]. The DSM-5 definition of probable AD (now called major neurocognitive disorder due to AD) differs from prior versions in that the cognitive domains have been renamed and expanded to include learning and memory, language, executive function, complex attention, perceptual-motor, and social cognition. Previously, the criteria recognized five domains (memory, aphasia, apraxia, agnosia, and executive function). Like prior versions, the criteria continue to require both memory impairment and evidence of decline in at least one other cognitive domain. New to the criteria is the recognition of genetic testing results, if known, as supportive of a diagnosis of probable AD.

While less substantially validated compared with the NIA-AA criteria, the DSM-IV criteria appeared to have similar accuracy [99,171]. The performance characteristics of the DSM-5 criteria compared with the DSM-IV and NIA-AA criteria are not yet known.

MCI due to Alzheimer disease — Amnestic mild cognitive impairment (MCI) refers to a state of circumscribed anterograde long-term memory impairment with preserved general cognitive and social functioning [172]. Amnestic MCI frequently represents an early stage of AD, with a progression rate to dementia at approximately 10 to 15 percent per year [172,173].

Nonamnestic MCI, in which cognitive domains outside of memory are affected, is less frequently associated with prodromal AD but may be in a significant minority of patients. (See

"Mild cognitive impairment: Epidemiology, pathology, and clinical assessment", section on 'Amnestic MCI' and "Mild cognitive impairment: Prognosis and treatment", section on 'Progression to dementia'.)

Formal clinical diagnostic criteria for MCI due to AD have been developed by a panel convened by the NIA-AA [174]. These include:

- A concern regarding cognition reported by the patient or informant or observed by the clinician
- Objective evidence of impairment in one or more cognitive domains that is not explained by age or education
- Preservation of independence in functional abilities
- Impairments do not meet criteria for dementia

MCI may occur as a prodrome to several neurodegenerative dementias, as well as non-neurodegenerative conditions (eg, depression, medicine effects). Therefore, the specific designation of MCI due to AD is used when a biomarker associated with AD (eg, cerebrospinal fluid [CSF] testing of beta-amyloid [A β], tau, and phospho-tau; amyloid imaging; or functional scan consistent with AD) is present in the affected subject (table 2). Using the NIA-AA criteria on a cohort of research subjects with MCI, individuals with cognitive impairment and both amyloid and neuronal injury biomarkers had an approximately 60 percent progression rate to AD dementia over three years [175]. By contrast, individuals with amnestic MCI and both negative amyloid and neuronal injury markers have a low progression rate: 12 percent over three years in one study [176].

The International Working Group (IWG) recognizes a similar clinical presentation that, when accompanied by biomarker evidence of AD, is referred to as prodromal AD [150,177]. The designation of prodromal AD has been associated with similar progression rates compared with the NIA-AA criteria [175].

Preclinical Alzheimer disease — Preclinical AD refers to the stage of AD in which the molecular pathology of AD is already present in the brain but is not yet clinically expressed. Individuals are by definition asymptomatic and cognitively normal. Preclinical AD is defined by biomarker measures and is only recommended for use in research, in the context of clinical trials for early intervention strategies and the longitudinal study of people at risk. According to the IWG, preclinical AD encompasses two groups [178]:

- Asymptomatic at risk for AD refers to cognitively normal individuals with evidence of AD molecular pathology by laboratory or imaging biomarkers. It is not currently known whether progression to symptomatic AD is inevitable in such individuals. Autopsy data

revealing that people may die with normal cognition and pathologic changes of AD sufficient to make a diagnosis suggest that symptomatic AD will not occur in every subject with amyloid in the brain.

- Presymptomatic AD also refers to cognitively normal and asymptomatic carriers of a dominantly inherited gene mutation that causes AD. Symptoms of AD will almost certainly develop in these individuals over a normal lifespan. (See ["Genetics of Alzheimer disease", section on 'Early-onset Alzheimer disease'](#).)

The NIA-AA framework proposes three stages of preclinical AD, recognizing that some individuals will not progress beyond stage 1 or stage 2, and that individuals in stage 3 are probably most likely to progress to MCI and AD dementia ( [figure 2](#)) [179]. As described above, a subsequent NIA-AA research framework was published in 2018 that further explicates the continuum of AD, with the disease being defined by amyloid, tau, and neurodegeneration (ATN) biomarkers in the context of the clinical stages from preclinical to MCI to dementia stages [152]. These are research criteria that have not yet been validated through longitudinal studies, and they are not currently recommended for use outside a research setting.

DIFFERENTIAL DIAGNOSIS

The most common disorders considered in the differential diagnosis of AD are vascular dementia and other neurodegenerative dementias. The two most common neurodegenerative dementias after AD are dementia with Lewy bodies (DLB) and frontotemporal dementia (FTD).

- Vascular dementia is caused by either ischemic or hemorrhagic strokes or small vessel cerebrovascular disease. Diagnosis is most specific if there is a stroke-like course of illness, neurologic signs of stroke on examination, and imaging evidence of cerebrovascular disease. However, the course of illness may appear smoothly progressive, and there may be no elementary neurologic signs. Cerebrovascular disease commonly co-occurs with AD. With increasing age, it is more common than not to find both AD and cerebrovascular disease in the brain of a patient with dementia. (See ["Etiology, clinical manifestations, and diagnosis of vascular dementia"](#).)
- DLB may be the second most common type of degenerative dementia after AD. Clinical features that help distinguish this from AD include prominent early appearance of visual hallucinations, along with parkinsonism, cognitive fluctuations, dysautonomia, rapid eye movement (REM) sleep behavior disorder, and neuroleptic sensitivity. (See ["Clinical features and diagnosis of dementia with Lewy bodies"](#).)

- FTD is a neuropathologically and clinically heterogeneous disorder characterized by focal degeneration of the frontal and/or temporal lobes. Early alteration of personality, social and emotional behavior, and executive functioning are prominent clinical characteristics of behavioral variant FTD. Primary progressive aphasia (PPA) is a form of FTD in which gradually progressive language impairment is the core feature early in the course. There are three major subtypes, with the semantic and nonfluent variants being associated usually with frontotemporal lobar degeneration (FTLD) pathologies (usually tau or TDP-43). PPA, particularly the logopenic variant, can also be a presentation of AD. (See ["Frontotemporal dementia: Clinical features and diagnosis"](#) and ["Atypical presentations"](#) above.)

SOCIETY GUIDELINE LINKS

Links to society and government-sponsored guidelines from selected countries and regions around the world are provided separately. (See ["Society guideline links: Cognitive impairment and dementia"](#).)

INFORMATION FOR PATIENTS

UpToDate offers two types of patient education materials, "The Basics" and "Beyond the Basics." The Basics patient education pieces are written in plain language, at the 5th to 6th grade reading level, and they answer the four or five key questions a patient might have about a given condition. These articles are best for patients who want a general overview and who prefer short, easy-to-read materials. Beyond the Basics patient education pieces are longer, more sophisticated, and more detailed. These articles are written at the 10th to 12th grade reading level and are best for patients who want in-depth information and are comfortable with some medical jargon.

Here are the patient education articles that are relevant to this topic. We encourage you to print or e-mail these topics to your patients. (You can also locate patient education articles on a variety of subjects by searching on "patient info" and the keyword(s) of interest.)

- Basics topics (see ["Patient education: Dementia \(including Alzheimer disease\) \(The Basics\)"](#) and ["Patient education: Tips for caregivers of people with Alzheimer disease \(The Basics\)"](#) and ["Patient education: Evaluating memory and thinking problems \(The Basics\)"](#))
- Beyond the Basics topics (see ["Patient education: Dementia \(including Alzheimer disease\) \(Beyond the Basics\)"](#))

SUMMARY AND RECOMMENDATIONS

- **Age at onset** – Alzheimer disease (AD) is usually a disease of older age. The incidence increases exponentially with age over 65 years.

Early-onset (onset of symptoms in a person younger than 65) AD is unusual, and in some but not all cases is familial. Familial early-onset AD makes up less than 1 percent of cases (several hundred families around the world) and often follows an autosomal-dominant pattern of inheritance. (See ['Age of onset'](#) above.)

- **Clinical features**

- **Cognitive deficits** – These appear and progress insidiously. Memory impairment, specifically loss of memory of recent events, is the most frequent feature of AD and is usually its first manifestation. Other cognitive deficits appear with or after the development of memory impairment. Executive dysfunction and impaired visuospatial skills tend to be affected early, while deficits in language function and behavioral symptoms often manifest later. (See ['Clinical features'](#) above.)
- **Neuropsychiatric and behavioral symptoms** – These are common in middle and late stages of AD but may occur early in the course in some patients. (See ['Behavioral and psychologic symptoms'](#) above and ["Management of neuropsychiatric symptoms of dementia"](#).)
- **Noncognitive neurologic deficits** – Pyramidal and extrapyramidal motor signs, myoclonus, and seizures can occur in late stages of AD but are uncommon in early and middle stages. (See ['Other signs and symptoms'](#) above.)
- **Atypical presentations** – These include a visual variant (posterior cortical atrophy), a variant with progressive aphasia, and a variant with progressive executive dysfunction as the predominant symptom. Atypical presentations are more common in younger people with AD. (See ['Atypical presentations'](#) above.)
- **Clinical course** – AD is inexorably progressive, but the rate of progression can vary. The average life expectancy after diagnosis has been reported to be between 8 and 10 years but may range from 3 to 20 years. (See ['Clinical course'](#) above.)
- **Diagnosis** – AD should be suspected in any older adult with insidious onset, progressive decline in memory, and at least one other cognitive domain leading to impaired

functioning. The diagnosis of AD is made in large part by this clinical assessment. (See ['Clinical assessment'](#) above.)

Neuropsychologic testing may provide confirmatory information and aid in patient management. A neuroimaging study should be obtained on every patient suspected of having AD. (See ['Role of neuropsychologic testing'](#) above and ['Neuroimaging'](#) above.)

In selected cases (eg, those with young age of onset or atypical presentations), other imaging or biomarker tests including 18-F fluorodeoxyglucose positron emission tomography (FDG-PET), cerebrospinal fluid (CSF) testing, or amyloid/tau PET may be helpful, although access and reimbursement for these tests may present challenges. If use of [aducanumab](#) is being considered, confirmation of amyloid status is necessary with either amyloid PET or CSF testing. (See ['Neuroimaging'](#) above and ['Role of biomarkers'](#) above.)

ACKNOWLEDGMENT

The UpToDate editorial staff acknowledges Thomas Grabowski, Jr, MD, who contributed to an earlier version of this topic review.

Use of UpToDate is subject to the [Terms of Use](#).

REFERENCES

1. Ballard C, Gauthier S, Corbett A, et al. Alzheimer's disease. *Lancet* 2011; 377:1019.
2. Braak H, Braak E. Frequency of stages of Alzheimer-related lesions in different age categories. *Neurobiol Aging* 1997; 18:351.
3. Ryman DC, Acosta-Baena N, Aisen PS, et al. Symptom onset in autosomal dominant Alzheimer disease: A systematic review and meta-analysis. *Neurology* 2014; 83:253.
4. Schupf N, Kapell D, Nightingale B, et al. Earlier onset of Alzheimer's disease in men with Down syndrome. *Neurology* 1998; 50:991.
5. Markowitsch HJ, Staniloiu A. Amnesic disorders. *Lancet* 2012; 380:1429.
6. Mortamais M, Ash JA, Harrison J, et al. Detecting cognitive changes in preclinical Alzheimer's disease: A review of its feasibility. *Alzheimers Dement* 2017; 13:468.
7. Scoville WB, Milner B. Loss of recent memory after bilateral hippocampal lesions. *J Neurol Neurosurg Psychiatry* 1957; 20:11.

8. Zola-Morgan S, Squire LR, Amaral DG. Human amnesia and the medial temporal region: Enduring memory impairment following a bilateral lesion limited to field CA1 of the hippocampus. *J Neurosci* 1986; 6:2950.
9. Peters F, Collette F, Degueldre C, et al. The neural correlates of verbal short-term memory in Alzheimer's disease: An fMRI study. *Brain* 2009; 132:1833.
10. Wagner M, Wolf S, Reischies FM, et al. Biomarker validation of a cued recall memory deficit in prodromal Alzheimer disease. *Neurology* 2012; 78:379.
11. Ahmed S, Mitchell J, Arnold R, et al. Memory complaints in mild cognitive impairment, worried well, and semantic dementia patients. *Alzheimer Dis Assoc Disord* 2008; 22:227.
12. McKhann GM, Knopman DS, Chertkow H, et al. The diagnosis of dementia due to Alzheimer's disease: Recommendations from the National Institute on Aging-Alzheimer's Association workgroups on diagnostic guidelines for Alzheimer's disease. *Alzheimers Dement* 2011; 7:263.
13. American Psychiatric Association. *Diagnostic and Statistical Manual of Mental Disorders, Fifth Edition (DSM-5)*, American Psychiatric Association, Arlington, VA 2013.
14. Stokholm J, Vogel A, Gade A, Waldemar G. Heterogeneity in executive impairment in patients with very mild Alzheimer's disease. *Dement Geriatr Cogn Disord* 2006; 22:54.
15. Harwood DG, Sultzer DL, Feil D, et al. Frontal lobe hypometabolism and impaired insight in Alzheimer disease. *Am J Geriatr Psychiatry* 2005; 13:934.
16. Barrett AM, Eslinger PJ, Ballentine NH, Heilman KM. Unawareness of cognitive deficit (cognitive anosognosia) in probable AD and control subjects. *Neurology* 2005; 64:693.
17. McDaniel KD, Edland SD, Heyman A. Relationship between level of insight and severity of dementia in Alzheimer disease. CERAD Clinical Investigators. Consortium to Establish a Registry for Alzheimer's Disease. *Alzheimer Dis Assoc Disord* 1995; 9:101.
18. Harwood DG, Sultzer DL, Wheatley MV. Impaired insight in Alzheimer disease: Association with cognitive deficits, psychiatric symptoms, and behavioral disturbances. *Neuropsychiatry Neuropsychol Behav Neurol* 2000; 13:83.
19. Mizrahi R, Starkstein SE, Jorge R, Robinson RG. Phenomenology and clinical correlates of delusions in Alzheimer disease. *Am J Geriatr Psychiatry* 2006; 14:573.
20. Parakh R, Roy E, Koo E, Black S. Pantomime and imitation of limb gestures in relation to the severity of Alzheimer's disease. *Brain Cogn* 2004; 55:272.
21. Kato M, Meguro K, Sato M, et al. Ideomotor apraxia in patients with Alzheimer disease: Why do they use their body parts as objects? *Neuropsychiatry Neuropsychol Behav Neurol* 2001; 14:45.

22. Giannakopoulos P, Duc M, Gold G, et al. Pathologic correlates of apraxia in Alzheimer disease. *Arch Neurol* 1998; 55:689.
23. Sarazin M, Stern Y, Berr C, et al. Neuropsychological predictors of dependency in patients with Alzheimer disease. *Neurology* 2005; 64:1027.
24. Rahayel S, Frasnelli J, Joubert S. The effect of Alzheimer's disease and Parkinson's disease on olfaction: A meta-analysis. *Behav Brain Res* 2012; 231:60.
25. Sun GH, Raji CA, Maceachern MP, Burke JF. Olfactory identification testing as a predictor of the development of Alzheimer's dementia: A systematic review. *Laryngoscope* 2012; 122:1455.
26. Bahar-Fuchs A, Moss S, Rowe C, Savage G. Olfactory performance in AD, aMCI, and healthy ageing: A unirhinal approach. *Chem Senses* 2010; 35:855.
27. Stamps JJ, Bartoshuk LM, Heilman KM. A brief olfactory test for Alzheimer's disease. *J Neurol Sci* 2013; 333:19.
28. Quarmley M, Moberg PJ, Mechanic-Hamilton D, et al. Odor Identification Screening Improves Diagnostic Classification in Incipient Alzheimer's Disease. *J Alzheimers Dis* 2017; 55:1497.
29. Ju YE, Lucey BP, Holtzman DM. Sleep and Alzheimer disease pathology--A bidirectional relationship. *Nat Rev Neurol* 2014; 10:115.
30. Ju YE, McLeland JS, Toedebusch CD, et al. Sleep quality and preclinical Alzheimer disease. *JAMA Neurol* 2013; 70:587.
31. Spira AP, Gamaldo AA, An Y, et al. Self-reported sleep and β -amyloid deposition in community-dwelling older adults. *JAMA Neurol* 2013; 70:1537.
32. Brown BM, Rainey-Smith SR, Villemagne VL, et al. The relationship between sleep quality and brain amyloid burden. *Sleep* 2016; 39:1063.
33. Hauser WA, Morris ML, Heston LL, Anderson VE. Seizures and myoclonus in patients with Alzheimer's disease. *Neurology* 1986; 36:1226.
34. McAreavey MJ, Ballinger BR, Fenton GW. Epileptic seizures in elderly patients with dementia. *Epilepsia* 1992; 33:657.
35. Romanelli MF, Morris JC, Ashkin K, Coben LA. Advanced Alzheimer's disease is a risk factor for late-onset seizures. *Arch Neurol* 1990; 47:847.
36. Scarmeas N, Honig LS, Choi H, et al. Seizures in Alzheimer disease: Who, when, and how common? *Arch Neurol* 2009; 66:992.
37. Irizarry MC, Jin S, He F, et al. Incidence of new-onset seizures in mild to moderate Alzheimer disease. *Arch Neurol* 2012; 69:368.

38. Zarea A, Charbonnier C, Rovelet-Lecrux A, et al. Seizures in dominantly inherited Alzheimer disease. *Neurology* 2016; 87:912.
39. Vessel KA, Tartaglia MC, Nygaard HB, et al. Epileptic activity in Alzheimer's disease: Causes and clinical relevance. *Lancet Neurol* 2017; 16:311.
40. Portet F, Scarmeas N, Cosentino S, et al. Extrapyrarnidal signs before and after diagnosis of incident Alzheimer disease in a prospective population study. *Arch Neurol* 2009; 66:1120.
41. McKhann G, Drachman D, Folstein M, et al. Clinical diagnosis of Alzheimer's disease: Report of the NINCDS-ADRDA Work Group under the auspices of Department of Health and Human Services Task Force on Alzheimer's Disease. *Neurology* 1984; 34:939.
42. Galton CJ, Patterson K, Xuereb JH, Hodges JR. Atypical and typical presentations of Alzheimer's disease: A clinical, neuropsychological, neuroimaging and pathological study of 13 cases. *Brain* 2000; 123 Pt 3:484.
43. Alladi S, Xuereb J, Bak T, et al. Focal cortical presentations of Alzheimer's disease. *Brain* 2007; 130:2636.
44. Harasty JA, Halliday GM, Xuereb J, et al. Cortical degeneration associated with phonologic and semantic language impairments in AD. *Neurology* 2001; 56:944.
45. Ng SY, Villemagne VL, Masters CL, Rowe CC. Evaluating atypical dementia syndromes using positron emission tomography with carbon 11 labeled Pittsburgh Compound B. *Arch Neurol* 2007; 64:1140.
46. Bokde AL, Pietrini P, Ibáñez V, et al. The effect of brain atrophy on cerebral hypometabolism in the visual variant of Alzheimer disease. *Arch Neurol* 2001; 58:480.
47. Whitwell JL, Dickson DW, Murray ME, et al. Neuroimaging correlates of pathologically defined subtypes of Alzheimer's disease: A case-control study. *Lancet Neurol* 2012; 11:868.
48. Ridgway GR, Lehmann M, Barnes J, et al. Early-onset Alzheimer disease clinical variants: Multivariate analyses of cortical thickness. *Neurology* 2012; 79:80.
49. Balasa M, Gelpi E, Antonell A, et al. Clinical features and APOE genotype of pathologically proven early-onset Alzheimer disease. *Neurology* 2011; 76:1720.
50. van der Flier WM, Pijnenburg YA, Fox NC, Scheltens P. Early-onset versus late-onset Alzheimer's disease: The case of the missing APOE ϵ 4 allele. *Lancet Neurol* 2011; 10:280.
51. Mendez MF, Mendez MA, Martin R, et al. Complex visual disturbances in Alzheimer's disease. *Neurology* 1990; 40:439.
52. Cogan DG. Visual disturbances with focal progressive dementing disease. *Am J Ophthalmol* 1985; 100:68.

53. Hof PR, Bouras C, Constantinidis J, Morrison JH. Selective disconnection of specific visual association pathways in cases of Alzheimer's disease presenting with Balint's syndrome. *J Neuropathol Exp Neurol* 1990; 49:168.
54. Levine DN, Lee JM, Fisher CM. The visual variant of Alzheimer's disease: A clinicopathologic case study. *Neurology* 1993; 43:305.
55. Renner JA, Burns JM, Hou CE, et al. Progressive posterior cortical dysfunction: A clinicopathologic series. *Neurology* 2004; 63:1175.
56. Andrade K, Kas A, Valabrègue R, et al. Visuospatial deficits in posterior cortical atrophy: Structural and functional correlates. *J Neurol Neurosurg Psychiatry* 2012; 83:860.
57. Crutch SJ, Schott JM, Rabinovici GD, et al. Consensus classification of posterior cortical atrophy. *Alzheimers Dement* 2017; 13:870.
58. Migliaccio R, Agosta F, Rascovsky K, et al. Clinical syndromes associated with posterior atrophy: Early age at onset AD spectrum. *Neurology* 2009; 73:1571.
59. Tang-Wai DF, Graff-Radford NR, Boeve BF, et al. Clinical, genetic, and neuropathologic characteristics of posterior cortical atrophy. *Neurology* 2004; 63:1168.
60. Whitwell JL, Jack CR Jr, Kantarci K, et al. Imaging correlates of posterior cortical atrophy. *Neurobiol Aging* 2007; 28:1051.
61. Ross SJ, Graham N, Stuart-Green L, et al. Progressive biparietal atrophy: An atypical presentation of Alzheimer's disease. *J Neurol Neurosurg Psychiatry* 1996; 61:388.
62. Crutch SJ, Lehmann M, Schott JM, et al. Posterior cortical atrophy. *Lancet Neurol* 2012; 11:170.
63. Caroppo P, Belin C, Grabli D, et al. Posterior cortical atrophy as an extreme phenotype of GRN mutations. *JAMA Neurol* 2015; 72:224.
64. Mitchell SB, Lucente D, Larvie M, et al. A 63-year-old man with progressive visual symptoms. *JAMA Neurol* 2017; 74:114.
65. Josephs KA, Whitwell JL, Duffy JR, et al. Progressive aphasia secondary to Alzheimer disease vs FTLD pathology. *Neurology* 2008; 70:25.
66. Knibb JA, Xuereb JH, Patterson K, Hodges JR. Clinical and pathological characterization of progressive aphasia. *Ann Neurol* 2006; 59:156.
67. Mesulam M, Wicklund A, Johnson N, et al. Alzheimer and frontotemporal pathology in subsets of primary progressive aphasia. *Ann Neurol* 2008; 63:709.
68. Gorno-Tempini ML, Dronkers NF, Rankin KP, et al. Cognition and anatomy in three variants of primary progressive aphasia. *Ann Neurol* 2004; 55:335.

69. Rabinovici GD, Jagust WJ, Furst AJ, et al. Abeta amyloid and glucose metabolism in three variants of primary progressive aphasia. *Ann Neurol* 2008; 64:388.
70. Deramecourt V, Lebert F, Debachy B, et al. Prediction of pathology in primary progressive language and speech disorders. *Neurology* 2010; 74:42.
71. Hu WT, McMillan C, Libon D, et al. Multimodal predictors for Alzheimer disease in nonfluent primary progressive aphasia. *Neurology* 2010; 75:595.
72. Gorno-Tempini ML, Hillis AE, Weintraub S, et al. Classification of primary progressive aphasia and its variants. *Neurology* 2011; 76:1006.
73. Bergeron D, Gorno-Tempini ML, Rabinovici GD, et al. Prevalence of amyloid- β pathology in distinct variants of primary progressive aphasia. *Ann Neurol* 2018; 84:729.
74. Mesulam MM, Dickerson BC, Sherman JC, et al. Case 1-2017. A 70-Year-Old Woman with Gradually Progressive Loss of Language. *N Engl J Med* 2017; 376:158.
75. Dickerson BC, Wolk DA, Alzheimer's Disease Neuroimaging Initiative. Dysexecutive versus amnesic phenotypes of very mild Alzheimer's disease are associated with distinct clinical, genetic and cortical thinning characteristics. *J Neurol Neurosurg Psychiatry* 2011; 82:45.
76. Blennerhassett R, Lillo P, Halliday GM, et al. Distribution of pathology in frontal variant Alzheimer's disease. *J Alzheimers Dis* 2014; 39:63.
77. Galasko D, Hansen LA, Katzman R, et al. Clinical-neuropathological correlations in Alzheimer's disease and related dementias. *Arch Neurol* 1994; 51:888.
78. Arvanitakis Z, Capuano AW, Leurgans SE, et al. Relation of cerebral vessel disease to Alzheimer's disease dementia and cognitive function in elderly people: a cross-sectional study. *Lancet Neurol* 2016; 15:934.
79. Schneider JA, Arvanitakis Z, Yu L, et al. Cognitive impairment, decline and fluctuations in older community-dwelling subjects with Lewy bodies. *Brain* 2012; 135:3005.
80. Nag S, Yu L, Capuano AW, et al. Hippocampal sclerosis and TDP-43 pathology in aging and Alzheimer disease. *Ann Neurol* 2015; 77:942.
81. Connor DJ, Salmon DP, Sandy TJ, et al. Cognitive profiles of autopsy-confirmed Lewy body variant vs pure Alzheimer disease. *Arch Neurol* 1998; 55:994.
82. Weiner MF, Hynan LS, Parikh B, et al. Can Alzheimer's disease and dementias with Lewy bodies be distinguished clinically? *J Geriatr Psychiatry Neurol* 2003; 16:245.
83. Merdes AR, Hansen LA, Jeste DV, et al. Influence of Alzheimer pathology on clinical diagnostic accuracy in dementia with Lewy bodies. *Neurology* 2003; 60:1586.
84. Clark CM, Sheppard L, Fillenbaum GG, et al. Variability in annual Mini-Mental State Examination score in patients with probable Alzheimer disease: A clinical perspective of

- data from the Consortium to Establish a Registry for Alzheimer's Disease. *Arch Neurol* 1999; 56:857.
85. Adak S, Illouz K, Gorman W, et al. Predicting the rate of cognitive decline in aging and early Alzheimer disease. *Neurology* 2004; 63:108.
 86. Nourhashémi F, Ousset PJ, Gillette-Guyonnet S, et al. A 2-year follow-up of 233 very mild (CDR 0.5) Alzheimer's disease patients (REAL.FR cohort). *Int J Geriatr Psychiatry* 2008; 23:460.
 87. Han L, Cole M, Bellavance F, et al. Tracking cognitive decline in Alzheimer's disease using the mini-mental state examination: A meta-analysis. *Int Psychogeriatr* 2000; 12:231.
 88. Schmidt C, Wolff M, Weitz M, et al. Rapidly progressive Alzheimer disease. *Arch Neurol* 2011; 68:1124.
 89. Bernick C, Cummings J, Raman R, et al. Age and rate of cognitive decline in Alzheimer disease: Implications for clinical trials. *Arch Neurol* 2012; 69:901.
 90. Peters ME, Schwartz S, Han D, et al. Neuropsychiatric symptoms as predictors of progression to severe Alzheimer's dementia and death: The Cache County Dementia Progression Study. *Am J Psychiatry* 2015; 172:460.
 91. Helzner EP, Scarmeas N, Cosentino S, et al. Survival in Alzheimer disease: A multiethnic, population-based study of incident cases. *Neurology* 2008; 71:1489.
 92. Larson EB, Shadlen MF, Wang L, et al. Survival after initial diagnosis of Alzheimer disease. *Ann Intern Med* 2004; 140:501.
 93. Wolfson C, Wolfson DB, Asgharian M, et al. A reevaluation of the duration of survival after the onset of dementia. *N Engl J Med* 2001; 344:1111.
 94. Rountree SD, Chan W, Pavlik VN, et al. Factors that influence survival in a probable Alzheimer disease cohort. *Alzheimers Res Ther* 2012; 4:16.
 95. Nasreddine ZS, Phillips NA, Bédirian V, et al. The Montreal Cognitive Assessment, MoCA: A brief screening tool for mild cognitive impairment. *J Am Geriatr Soc* 2005; 53:695.
 96. Davis DH, Creavin ST, Yip JL, et al. Montreal Cognitive Assessment for the diagnosis of Alzheimer's disease and other dementias. *Cochrane Database Syst Rev* 2015; :CD010775.
 97. Rossetti HC, Lacritz LH, Cullum CM, Weiner MF. Normative data for the Montreal Cognitive Assessment (MoCA) in a population-based sample. *Neurology* 2011; 77:1272.
 98. Shaughnessy L, Sheard S, Goldfarb D, Atri A. Cognitive assessment of Alzheimer's disease and dementias in clinical practice: Pragmatics of brief instruments and neuropsychological evaluation. *J Clin Psychiatry* 2019; 80.

99. Knopman DS, DeKosky ST, Cummings JL, et al. Practice parameter: Diagnosis of dementia (an evidence-based review). Report of the Quality Standards Subcommittee of the American Academy of Neurology. *Neurology* 2001; 56:1143.
100. van de Pol LA, Hensel A, Barkhof F, et al. Hippocampal atrophy in Alzheimer disease: Age matters. *Neurology* 2006; 66:236.
101. Barkhof F, Polvikoski TM, van Straaten EC, et al. The significance of medial temporal lobe atrophy: A postmortem MRI study in the very old. *Neurology* 2007; 69:1521.
102. Whitwell JL, Josephs KA, Murray ME, et al. MRI correlates of neurofibrillary tangle pathology at autopsy: A voxel-based morphometry study. *Neurology* 2008; 71:743.
103. Kantarci K, Avula R, Senjem ML, et al. Dementia with Lewy bodies and Alzheimer disease: Neurodegenerative patterns characterized by DTI. *Neurology* 2010; 74:1814.
104. Raji CA, Lopez OL, Kuller LH, et al. Age, Alzheimer disease, and brain structure. *Neurology* 2009; 73:1899.
105. Wahlund LO, Almkvist O, Blennow K, et al. Evidence-based evaluation of magnetic resonance imaging as a diagnostic tool in dementia workup. *Top Magn Reson Imaging* 2005; 16:427.
106. Dickerson BC, Wolk DA, Alzheimer's Disease Neuroimaging Initiative. MRI cortical thickness biomarker predicts AD-like CSF and cognitive decline in normal adults. *Neurology* 2012; 78:84.
107. McEvoy LK, Holland D, Hagler DJ Jr, et al. Mild cognitive impairment: Baseline and longitudinal structural MR imaging measures improve predictive prognosis. *Radiology* 2011; 259:834.
108. Minoshima S, Giordani B, Berent S, et al. Metabolic reduction in the posterior cingulate cortex in very early Alzheimer's disease. *Ann Neurol* 1997; 42:85.
109. Silverman DH, Small GW, Chang CY, et al. Positron emission tomography in evaluation of dementia: Regional brain metabolism and long-term outcome. *JAMA* 2001; 286:2120.
110. Powers WJ, Perlmuter JS, Videen TO, et al. Blinded clinical evaluation of positron emission tomography for diagnosis of probable Alzheimer's disease. *Neurology* 1992; 42:765.
111. Duara R, Grady C, Haxby J, et al. Positron emission tomography in Alzheimer's disease. *Neurology* 1986; 36:879.
112. O'Brien JL, O'Keefe KM, LaViolette PS, et al. Longitudinal fMRI in elderly reveals loss of hippocampal activation with clinical decline. *Neurology* 2010; 74:1969.
113. Hu WT, Wang Z, Lee VM, et al. Distinct cerebral perfusion patterns in FTLN and AD. *Neurology* 2010; 75:881.

114. Pickut BA, Saerens J, Mariën P, et al. Discriminative use of SPECT in frontal lobe-type dementia versus (senile) dementia of the Alzheimer's type. *J Nucl Med* 1997; 38:929.
115. Ishii K, Sakamoto S, Sasaki M, et al. Cerebral glucose metabolism in patients with frontotemporal dementia. *J Nucl Med* 1998; 39:1875.
116. Foster NL, Heidebrink JL, Clark CM, et al. FDG-PET improves accuracy in distinguishing frontotemporal dementia and Alzheimer's disease. *Brain* 2007; 130:2616.
117. Rabinovici GD, Rosen HJ, Alkalay A, et al. Amyloid vs FDG-PET in the differential diagnosis of AD and FTLT. *Neurology* 2011; 77:2034.
118. Nordberg A. PET imaging of amyloid in Alzheimer's disease. *Lancet Neurol* 2004; 3:519.
119. Rabinovici GD, Furst AJ, O'Neil JP, et al. 11C-PIB PET imaging in Alzheimer disease and frontotemporal lobar degeneration. *Neurology* 2007; 68:1205.
120. Edison P, Archer HA, Hinz R, et al. Amyloid, hypometabolism, and cognition in Alzheimer disease: An [11C]PIB and [18F]FDG PET study. *Neurology* 2007; 68:501.
121. Jack CR Jr, Lowe VJ, Senjem ML, et al. 11C PiB and structural MRI provide complementary information in imaging of Alzheimer's disease and amnestic mild cognitive impairment. *Brain* 2008; 131:665.
122. Rowe CC, Ackerman U, Browne W, et al. Imaging of amyloid beta in Alzheimer's disease with 18F-BAY94-9172, a novel PET tracer: Proof of mechanism. *Lancet Neurol* 2008; 7:129.
123. Small GW, Bookheimer SY, Thompson PM, et al. Current and future uses of neuroimaging for cognitively impaired patients. *Lancet Neurol* 2008; 7:161.
124. Ances BM, Christensen JJ, Teshome M, et al. Cognitively unimpaired HIV-positive subjects do not have increased 11C-PiB: A case-control study. *Neurology* 2010; 75:111.
125. Roe CM, Mintun MA, Ghoshal N, et al. Alzheimer disease identification using amyloid imaging and reserve variables: Proof of concept. *Neurology* 2010; 75:42.
126. Vandenberghe R, Van Laere K, Ivanoiu A, et al. 18F-flutemetamol amyloid imaging in Alzheimer disease and mild cognitive impairment: A phase 2 trial. *Ann Neurol* 2010; 68:319.
127. Tolboom N, van der Flier WM, Boverhoff J, et al. Molecular imaging in the diagnosis of Alzheimer's disease: Visual assessment of [11C]PIB and [18F]FDDNP PET images. *J Neurol Neurosurg Psychiatry* 2010; 81:882.
128. Clark CM, Schneider JA, Bedell BJ, et al. Use of florbetapir-PET for imaging beta-amyloid pathology. *JAMA* 2011; 305:275.
129. Barthel H, Gertz HJ, Dresel S, et al. Cerebral amyloid- β PET with florbetaben (18F) in patients with Alzheimer's disease and healthy controls: A multicentre phase 2 diagnostic study. *Lancet Neurol* 2011; 10:424.

130. Wolk DA, Grachev ID, Buckley C, et al. Association between in vivo fluorine 18-labeled flutemetamol amyloid positron emission tomography imaging and in vivo cerebral cortical histopathology. *Arch Neurol* 2011; 68:1398.
131. Fleisher AS, Chen K, Liu X, et al. Using positron emission tomography and florbetapir F18 to image cortical amyloid in patients with mild cognitive impairment or dementia due to Alzheimer disease. *Arch Neurol* 2011; 68:1404.
132. Wolk DA, Zhang Z, Boudhar S, et al. Amyloid imaging in Alzheimer's disease: Comparison of florbetapir and Pittsburgh compound-B positron emission tomography. *J Neurol Neurosurg Psychiatry* 2012; 83:923.
133. Clark CM, Pontecorvo MJ, Beach TG, et al. Cerebral PET with florbetapir compared with neuropathology at autopsy for detection of neuritic amyloid- β plaques: A prospective cohort study. *Lancet Neurol* 2012; 11:669.
134. Doraiswamy PM, Sperling RA, Coleman RE, et al. Amyloid- β assessed by florbetapir F 18 PET and 18-month cognitive decline: A multicenter study. *Neurology* 2012; 79:1636.
135. Lim YY, Ellis KA, Pietrzak RH, et al. Stronger effect of amyloid load than APOE genotype on cognitive decline in healthy older adults. *Neurology* 2012; 79:1645.
136. Niedowicz DM, Beckett TL, Matveev S, et al. Pittsburgh compound B and the postmortem diagnosis of Alzheimer disease. *Ann Neurol* 2012; 72:564.
137. Ikonomic MD, Klunk WE, Abrahamson EE, et al. Post-mortem correlates of in vivo PiB-PET amyloid imaging in a typical case of Alzheimer's disease. *Brain* 2008; 131:1630.
138. Ossenkoppele R, Jansen WJ, Rabinovici GD, et al. Prevalence of amyloid PET positivity in dementia syndromes: A meta-analysis. *JAMA* 2015; 313:1939.
139. Jansen WJ, Ossenkoppele R, Knol DL, et al. Prevalence of cerebral amyloid pathology in persons without dementia: A meta-analysis. *JAMA* 2015; 313:1924.
140. Boccardi M, Altomare D, Ferrari C, et al. Assessment of the incremental diagnostic value of florbetapir F 18 imaging in patients with cognitive impairment: The incremental diagnostic value of amyloid PET with [18F]-florbetapir (INDIA-FBP) study. *JAMA Neurol* 2016; 73:1417.
141. Rabinovici GD, Gatsonis C, Apgar C, et al. Association of Amyloid Positron Emission Tomography With Subsequent Change in Clinical Management Among Medicare Beneficiaries With Mild Cognitive Impairment or Dementia. *JAMA* 2019; 321:1286.
142. Yang L, Rieves D, Ganley C. Brain amyloid imaging--FDA approval of florbetapir F18 injection. *N Engl J Med* 2012; 367:885.
143. Drugs@FDA. <http://www.accessdata.fda.gov/scripts/cder/drugsatfda/index.cfm> (Accessed on December 11, 2020).

144. Johnson KA, Minoshima S, Bohnen NI, et al. Appropriate use criteria for amyloid PET: A report of the Amyloid Imaging Task Force, the Society of Nuclear Medicine and Molecular Imaging, and the Alzheimer's Association. *J Nucl Med* 2013; 54:476.
145. Cummings J, Salloway S. Aducanumab: Appropriate use recommendations. *Alzheimers Dement* 2022; 18:531.
146. Xia C, Makaretz SJ, Caso C, et al. Association of In Vivo [18F]AV-1451 Tau PET Imaging Results With Cortical Atrophy and Symptoms in Typical and Atypical Alzheimer Disease. *JAMA Neurol* 2017; 74:427.
147. Villemagne VL, Okamura N, Rowe CC. Untangling tau imaging. *Alzheimers Dement (Amst)* 2016; 4:39.
148. Fleisher AS, Pontecorvo MJ, Devous MD Sr, et al. Positron Emission Tomography Imaging With [18F]flortaucipir and Postmortem Assessment of Alzheimer Disease Neuropathologic Changes. *JAMA Neurol* 2020; 77:829.
149. Ghirelli A, Tosakulwong N, Weigand SD, et al. Sensitivity-Specificity of Tau and Amyloid β Positron Emission Tomography in Frontotemporal Lobar Degeneration. *Ann Neurol* 2020; 88:1009.
150. Dubois B, Feldman HH, Jacova C, et al. Revising the definition of Alzheimer's disease: A new lexicon. *Lancet Neurol* 2010; 9:1118.
151. Wolk DA, Sadowsky C, Safirstein B, et al. Use of flutemetamol F 18-labeled positron emission tomography and other biomarkers to assess risk of clinical progression in patients with amnesic mild cognitive impairment. *JAMA Neurol* 2018; 75:1114.
152. Jack CR Jr, Bennett DA, Blennow K, et al. NIA-AA Research Framework: Toward a biological definition of Alzheimer's disease. *Alzheimers Dement* 2018; 14:535.
153. Galasko D. Cerebrospinal fluid biomarkers in Alzheimer disease: A fractional improvement? *Arch Neurol* 2003; 60:1195.
154. Bateman RJ, Wen G, Morris JC, Holtzman DM. Fluctuations of CSF amyloid-beta levels: Implications for a diagnostic and therapeutic biomarker. *Neurology* 2007; 68:666.
155. Bouwman FH, van der Flier WM, Schoonenboom NS, et al. Longitudinal changes of CSF biomarkers in memory clinic patients. *Neurology* 2007; 69:1006.
156. Sonnen JA, Montine KS, Quinn JF, et al. Biomarkers for cognitive impairment and dementia in elderly people. *Lancet Neurol* 2008; 7:704.
157. Bateman RJ, Xiong C, Benzinger TL, et al. Clinical and biomarker changes in dominantly inherited Alzheimer's disease. *N Engl J Med* 2012; 367:795.

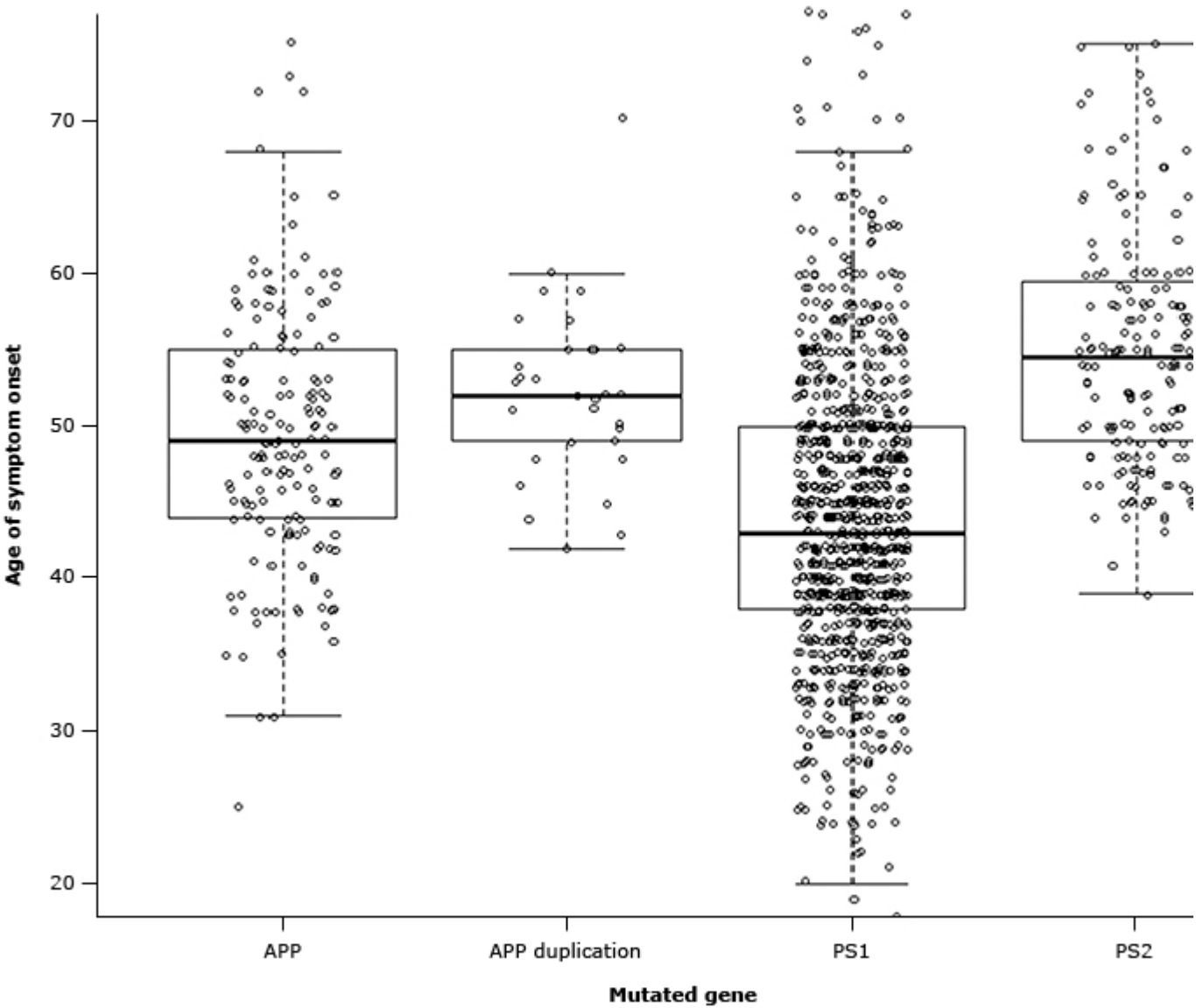
158. Doecke JD, Laws SM, Faux NG, et al. Blood-based protein biomarkers for diagnosis of Alzheimer disease. *Arch Neurol* 2012; 69:1318.
159. Rasmussen KL, Tybjaerg-Hansen A, Nordestgaard BG, Frikke-Schmidt R. Plasma levels of apolipoprotein E and risk of dementia in the general population. *Ann Neurol* 2015; 77:301.
160. Olsson B, Lautner R, Andreasson U, et al. CSF and blood biomarkers for the diagnosis of Alzheimer's disease: A systematic review and meta-analysis. *Lancet Neurol* 2016; 15:673.
161. Karikari TK, Pascoal TA, Ashton NJ, et al. Blood phosphorylated tau 181 as a biomarker for Alzheimer's disease: A diagnostic performance and prediction modelling study using data from four prospective cohorts. *Lancet Neurol* 2020; 19:422.
162. Palmqvist S, Janelidze S, Quiroz YT, et al. Discriminative Accuracy of Plasma Phospho-tau217 for Alzheimer Disease vs Other Neurodegenerative Disorders. *JAMA* 2020; 324:772.
163. Mattsson-Carlgrén N, Janelidze S, Palmqvist S, et al. Longitudinal plasma p-tau217 is increased in early stages of Alzheimer's disease. *Brain* 2020; 143:3234.
164. Schindler SE, Bollinger JG, Ovod V, et al. High-precision plasma β -amyloid 42/40 predicts current and future brain amyloidosis. *Neurology* 2019; 93:e1647.
165. Mayeux R, Saunders AM, Shea S, et al. Utility of the apolipoprotein E genotype in the diagnosis of Alzheimer's disease. Alzheimer's Disease Centers Consortium on Apolipoprotein E and Alzheimer's Disease. *N Engl J Med* 1998; 338:506.
166. Petersen RC, Smith GE, Ivnik RJ, et al. Apolipoprotein E status as a predictor of the development of Alzheimer's disease in memory-impaired individuals. *JAMA* 1995; 273:1274.
167. Braak H, Braak E. Neuropathological staging of Alzheimer-related changes. *Acta Neuropathol* 1991; 82:239.
168. Consensus recommendations for the postmortem diagnosis of Alzheimer's disease. The National Institute on Aging, and Reagan Institute Working Group on Diagnostic Criteria for the Neuropathological Assessment of Alzheimer's Disease. *Neurobiol Aging* 1997; 18:S1.
169. Khachaturian ZS. Diagnosis of Alzheimer's disease. *Arch Neurol* 1985; 42:1097.
170. American Psychiatric Association. Diagnostic and Statistical Manual of Mental Disorders, Fifth Edition (DSM-5), American Psychiatric Association, Arlington, VA 2013.
171. Phung TK, Andersen BB, Høgh P, et al. Validity of dementia diagnoses in the Danish hospital registers. *Dement Geriatr Cogn Disord* 2007; 24:220.
172. Petersen RC, Smith GE, Waring SC, et al. Mild cognitive impairment: Clinical characterization and outcome. *Arch Neurol* 1999; 56:303.
173. Tierney MC, Szalai JP, Snow WG, et al. Prediction of probable Alzheimer's disease in memory-impaired patients: A prospective longitudinal study. *Neurology* 1996; 46:661.

174. Albert MS, DeKosky ST, Dickson D, et al. The diagnosis of mild cognitive impairment due to Alzheimer's disease: Recommendations from the National Institute on Aging-Alzheimer's Association workgroups on diagnostic guidelines for Alzheimer's disease. *Alzheimers Dement* 2011; 7:270.
175. Vos SJ, Verhey F, Frölich L, et al. Prevalence and prognosis of Alzheimer's disease at the mild cognitive impairment stage. *Brain* 2015; 138:1327.
176. Wolk D, Salloway S, Dickerson B. Putting the New Alzheimer Disease Amyloid, Tau, Neurodegeneration (AT[N]) Diagnostic System to the Test. *JAMA* 2019; 321:2289.
177. Dubois B, Feldman HH, Jacova C, et al. Advancing research diagnostic criteria for Alzheimer's disease: The IWG-2 criteria. *Lancet Neurol* 2014; 13:614.
178. Morris JC, Blennow K, Froelich L, et al. Harmonized diagnostic criteria for Alzheimer's disease: Recommendations. *J Intern Med* 2014; 275:204.
179. Sperling RA, Aisen PS, Beckett LA, et al. Toward defining the preclinical stages of Alzheimer's disease: Recommendations from the National Institute on Aging-Alzheimer's Association workgroups on diagnostic guidelines for Alzheimer's disease. *Alzheimers Dement* 2011; 7:280.

Topic 5071 Version 39.0

GRAPHICS

Age at symptom onset by mutated gene in autosomal dominant Alzheimer disease



Ages at symptom onset for all affected individuals are shown grouped by the affected gene, with *APP* mutant having an additional subclass for gene duplications.

APP: amyloid precursor protein; PS1: presenilin 1; PS2: presenilin 2.

From: Ryman D, Acosta-Baena N, Aisen P, et al. Symptom onset in autosomal dominant Alzheimer disease: A systematic review and meta-analysis. *Neurology* 2014; 83:253. DOI: [10.1212/WNL.0000000000000596](https://doi.org/10.1212/WNL.0000000000000596). Reproduced with permission from Lippincott Williams & Wilkins. Copyright © 2014 American Academy of Neurology. Unauthorized reproduction of this material is prohibited.

Clinical dementia rating (CDR): 0, 0.5, 1, 2, 3

Impairment	None (0)	Questionable (0.5)	Mild (1)	Moderate (2)	Severe (3)
Memory	No memory loss or slight inconstant forgetfulness	Consistent slight forgetfulness; partial recollection of events	Moderate memory loss; more marked for recent events; defect interferes with everyday activities	Severe memory loss; only highly learned material retained; new material rapidly lost	Severe memory loss; only fragments remain
Orientation	Fully oriented	Fully oriented or slight difficulty with time relationships	Moderate difficulty with time relationships; oriented for place at examination; may have geographic disorientation elsewhere	Severe difficulty with time relationships; usually disoriented in time, often to place	Oriented to person only
Judgment and problem	Solves everyday problems and handles business and financial affairs well; judgment good in relation to past performance	Slight impairment to solving problems, similarities, differences	Moderate difficulty in handling problems, similarities, differences; social judgment usually maintained	Severely impaired in handling problems, similarities, differences; social judgment usually impaired	Unable to make judgments or solve problems
Community affairs	Independent function at usual level in job, shopping, volunteer and social groups	Slight impairment in these activities	Unable to function independently at these activities though may still be engaged in	No pretense of independent function outside of home; appears well enough to be	No pretense of independent function outside of home; appears too ill to be

			some; appears normal to casual inspection	taken to functions outside of family home	taken to functions outside a family home
Home and hobbies	Life at home, hobbies, intellectual interests well maintained	Life at home, hobbies, intellectual interests slightly impaired	Mild but definite impairment of function at home; more difficult chores abandoned; more complicated hobbies and interests abandoned	Only simple chores preserved; very restricted interests, poorly maintained	No significant function in home
Personal care	Fully capable of self care	Fully capable of self care	Needs prompting	Requires assistance in dressing, hygiene, keeping of personal effects	Requires much help with personal care; frequent incontinence

Score only as decline from previous usual level due to cognitive loss, not impaired due to other factors.

Reproduced with permission from: Morris JC. The clinical dementia rating (CDR): Current version and scoring rules. Neurology 1993; 43:2412. Copyright © 1993 Lippincott Williams & Wilkins.

Graphic 53706 Version 13.0

Spectrum of widely investigated biomarkers for Alzheimer disease

	Pathophysiologic marker	Topographic marker	Marker of neuronal injury/degeneration
Cerebrospinal fluid			
A β 42 or A β 42:A β 40 ratio	Yes	No	No
Total tau, phospho-tau	Yes	No	Yes
PET			
Amyloid tracer uptake	Yes	No	No
Fluorodeoxyglucose	No	Yes	Yes
Structural MRI			
Medial temporal lobe atrophy	No	Yes	Yes

A β : beta-amyloid; MRI: magnetic resonance imaging.

Graphic 97451 Version 2.0

DSM-5 criteria for major neurocognitive disorder due to Alzheimer disease

A. Evidence of significant cognitive decline from a previous level of performance in one or more cognitive domains*:
Learning and memory.
Language.
Executive function.
Complex attention.
Perceptual-motor.
Social cognition.
B. The cognitive deficits interfere with independence in everyday activities. At a minimum, assistance should be required with complex instrumental activities of daily living, such as paying bills or managing medications.
C. The cognitive deficits do not occur exclusively in the context of a delirium.
D. The cognitive deficits are not better explained by another mental disorder (eg, major depressive disorder, schizophrenia).
E. There is insidious onset and gradual progression of impairment in at least two cognitive domains.
F. Either of the following:
Evidence of a causative Alzheimer disease genetic mutation from family history or genetic testing.
All three of the following are present:
1) Clear evidence of decline in memory and learning and at least one other cognitive domain.
2) Steadily progressive, gradual decline in cognition, without extended plateaus.
3) No evidence of mixed etiology (ie, absence of other neurodegenerative disorders or cerebrovascular disease, or another neurological, mental, or systemic disease or condition likely contributing to cognitive decline).

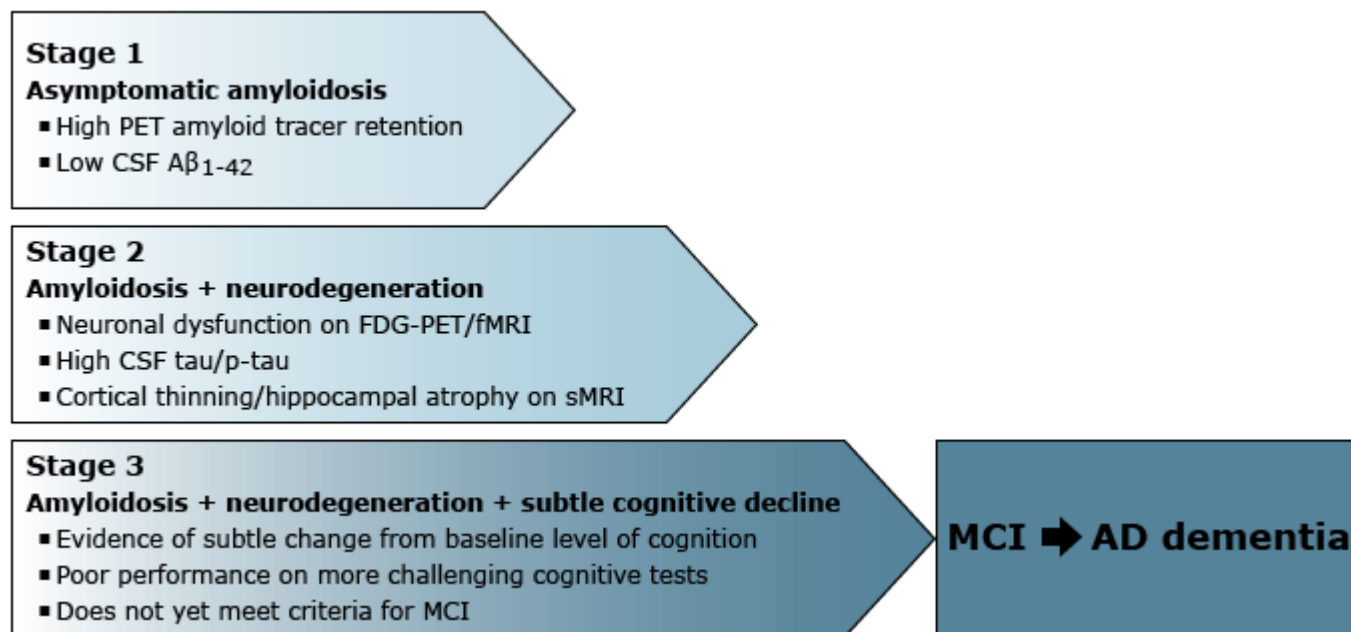
DSM: diagnostic and statistical manual.

* Evidence of decline is based on: Concern of the individual, a knowledgeable informant, or the clinician that there has been a significant decline in cognitive function; and a substantial impairment in cognitive performance, preferably documented by standardized neuropsychological testing or, in its absence, another quantified clinical assessment.

American Psychiatric Association. Diagnostic and Statistical Manual of Mental Disorders, Fifth Edition (DSM-5), American Psychiatric Association, Arlington, VA 2013.

Graphic 91288 Version 3.0

Proposed staging framework for preclinical Alzheimer disease



Graphic representation of the proposed staging framework for preclinical AD. Note that some individuals will not progress beyond Stage 1 or Stage 2. Individuals in Stage 3 are postulated to be more likely to progress to MCI and AD dementia.

AD: Alzheimer's disease; A β : amyloid beta; PET: position emission tomography; CSF: cerebrospinal fluid; FDG: fluorodeoxyglucose; fMRI: functional magnetic resonance imaging; sMRI: structural magnetic resonance imaging.

*Reproduced from: Sperling RA, Aisen PS, Beckett LA, et al. Toward defining the preclinical stages of Alzheimer's disease: recommendations from the National Institute on Aging-Alzheimer's Association workgroups on diagnostic guidelines for Alzheimer's disease. *Alzheimers Dement* 2011; 7:280. Illustration used with the permission of Elsevier Inc. All rights reserved.*

Graphic 97452 Version 2.0

